8. $\frac{dy}{dx}+\frac{y}{x}=xe^{x}$

9. A bank pays interest by continuous compounding , that is by treating the interest rate as the instantaneous rate of change of principal. A man invests ₹1,00,000 in the bank deposit which accures interest, 8% per year compounded continuously. How much will he get after 10 years.

4.3 Second Order first degree differential equations with constant coefficients:

4.3.1 A general second order linear differential equation with constant coefficients is of the form

$$a\frac{d^2y}{dx^2}+b\frac{dy}{dx}+cy=f(x)$$

$aD^2y+bDy+cy=f(x)$, where $\frac{d}{dx}=D, \frac{d^2}{dx^2}=D^2$

$$\phi(D)y=f(x) \qquad (1)$$

where $\phi(D)=aD^2+bD+c$ (a,b and c are constants)

To solve the equation (1), we first solve the equation $\phi(D)y=0$. The solution so obtained is called complementary function (*C.F*).

Next we operate on $f(x)$ with $\frac{1}{\phi(D)}$, the solution so obtained is called particular integral (*P.I*)

$$PI=\frac{1}{\phi(D)}f(x)$$

General solution is $y=C.F+P.I$

Type 1 : $f(x)=0$

$(i.e)\ \phi(D)y=0$

To solve this, put $\phi(D)=0$

Replace D by m. This equation is called auxiliary equation $.\phi(m)=0$ *is a* quadratic equation. So we have two roots, say m_1 and m_2.

Now we have the following three cases.

	Nature of roots	Complementary function
1	Real and different $(m_1\neq m_2)$	$Ae^{m_1x}+Be^{m_2x}$
2	Real and equal $m_1=m_2=m(say)$	$(Ax+B)e^{mx}$
3	Complex roots $(\alpha\pm i\beta)$	$e^{\alpha x}(A\cos\beta x+B\sin\beta x)$

Here A and B are arbitrary constants

Example 4.25

Solve $(D^2-3D-4)y=0$

Solution:

Given $(D^2-3D-4)y = 0$

The auxiliary equations is

$$m^2-3m-4 = 0$$
$$\Rightarrow (m-4)(m+1) = 0$$
$$m = -1, 4$$

Roots are real and different

$\therefore$ The complementary function is $Ae^{-x}+Be^{4x}$

The general solution is $y = Ae^{-x}+Be^{4x}$

Example 4.26

Solve $9y''-12y'+4y=0$

Solution:

Given $(9D^2-12D+4)y = 0$

The auxiliary equation is $(3m-2)^2=0$

$(3m-2)(3m-2)=0 \Rightarrow m=\frac{2}{3},\frac{2}{3}$

Roots are real and equal.

XII Std - Business Maths & Stat EM Chapter 4.indd 95 2/28/2020 6:02:54 PM